

Preferences toward risk, Insurance

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Insurance

- People can protect their wealth (W) against risky events (lotteries) purchasing insurance policies
- An ***insurance policy*** is a contract that, at a price, reduces the financial loss associated with a risky event:
 - It specifies a **BENEFIT** (B), the amount the policyholder receives if the loss occurs, and a **PREMIUM** (P), the price of the insurance.
 - Full coverage: it covers the whole wealth ($B=W$) (we look at this case)
 - Partial coverage: it covers only in part (we do not study it)
- We assume there are only two possible outcomes of a risky event: a bad outcome (loss) with probability p and a good outcome (no loss) with probability $(1-p)$

Insurance

- Full insurance coverage turns a lottery into a riskless lottery. Having paid the premium:
 - if a loss occurs, the policy holder receives the benefit, for a net gain of
$$\textit{Benefit} - \textit{Premium}$$
 - If a loss does not occur the consumer keeps the object:
$$\textit{Value of the object}(=\textit{Benefit}) - \textit{Premium}$$
- Agents under full insurance end up having *with certainty* a wealth lower than the initial one, W .

Example

- Giovanni owns a painting which he values at $W=900$ euros
 - The probability of the painting being stolen is: $p=0.5$
 - This situation can be interpreted as a lottery:
 - With probability 0.5 Giovanni keeps the painting and gets 900 euros
 - With probability 0.5 Giovanni gets robbed and ends up with 0 euros
 - If Giovanni decides to fully insure the painting and pays a premium P :
 - in case of a robbery he has $900-P$ euros
 - In case of no robbery he has $900-P$ euros
- ➔ the insurance is turning the lottery into a riskless one

Insurance

- Would agents be willing to insure themselves?
How much would they be willing to pay?
- The answer to these questions depends on two things:
 - the size of the premium asked by the insurance company
 - the agent's risk preferences

Company side: size of the premium

- How much would an insurance company ask in order to insure an agent?
 - A company decides to offer an insurance policy only if it makes non-negative expected profit (risk neutrality)
- The expected profits of a company are given by:

$$E\Pi = P - pB - (1 - p)0$$

where P is the premium,

B is the benefit and

p is the probability of the bad outcome

Company side: size of the premium

- A company decides to offer an insurance policy if $P \geq p_B$
- If $P = p_B$, the insurance is called *actuarially fair* and the premium is a *fair premium*.
- If $P > p_B$, the premium is called *actuarially unfair*.

Agents' side

- Would agents be willing to be insured for a premium $P \geq pB$?
- In order to answer we need to compare expected utilities with and without insurance.

$$EU(\text{no insurance}) = pu(0) + (1-p)u(W)$$

$$U(\text{insurance}) = pu(W-P) + (1-p)u(W-P) = u(W-P)$$

- Then, an agent is willing to buy full coverage if and only if:

$$U(\text{insurance}) \geq EU(\text{no insurance})$$

- Let us first compute the expected value of the lottery with and without insurance and then discuss the point for each type of agent, according to his preferences toward risk.

Expected value

- $EV(\text{no insurance}) = p0 + (1-p)W = (1-p)W$
- $EV(\text{insurance}) = p(W-P) + (1-p)(W-P) = W-P$
- $EV(\text{fair insurance}) = W - pW = (1-p)W$
 - Under an actuarially fair insurance agents get for sure an amount equal to the expected value of the lottery under no insurance.
- $EV(\text{unfair insurance}) = W - P < (1-p)W$
 - Under a less than actuarially fair agents get for sure an amount lower than the expected value under no insurance.

Risk-loving and risk-neutral agents

- If a risk loving agent is offered an actuarially fair contract, he will never choose full coverage
 - $EV(\text{fair insurance}) = EV(\text{no insurance})$
 - ➔ $U(\text{fair insurance}) < EU(\text{no insurance})$

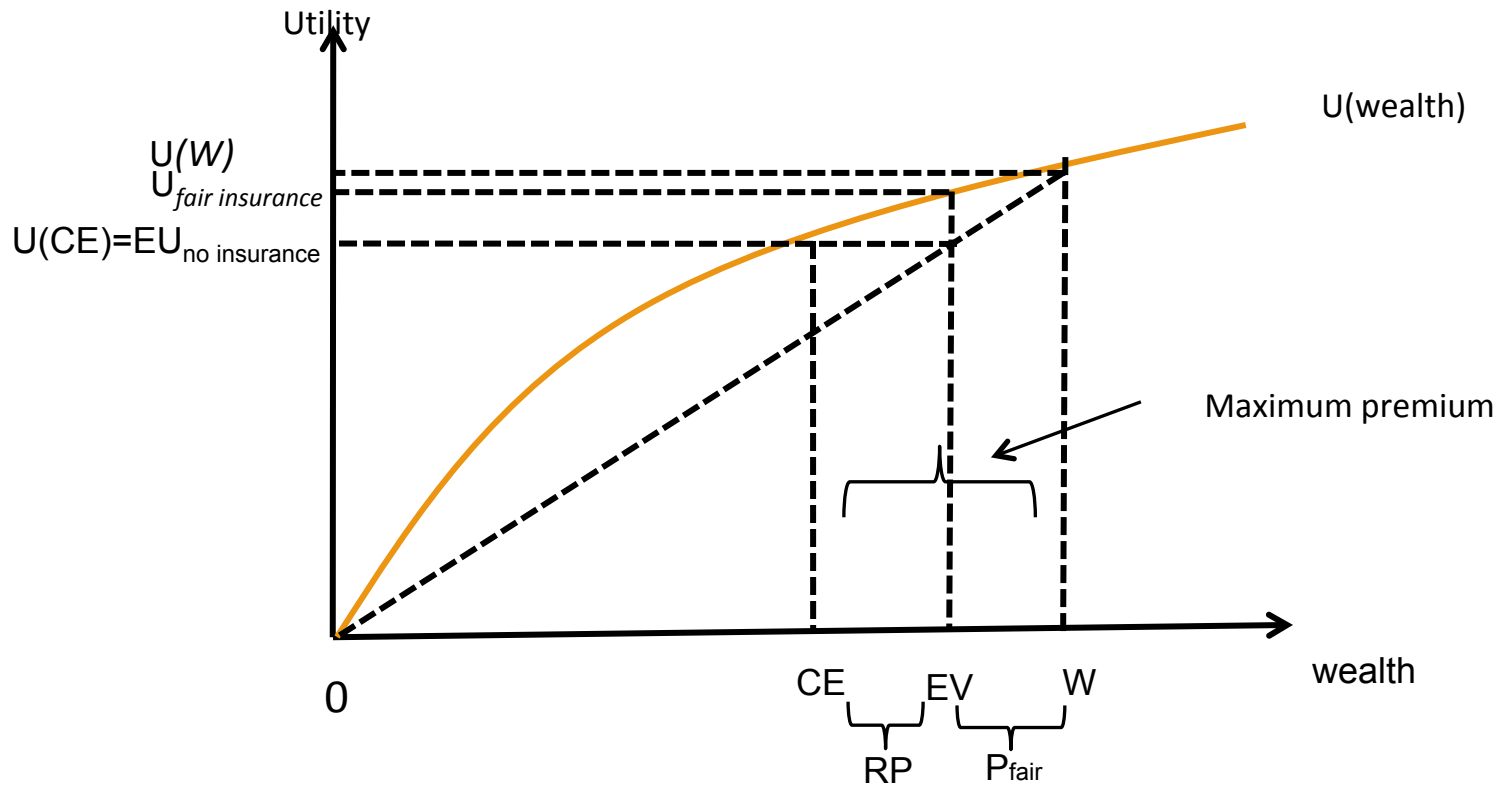
- If a risk neutral agent is offered an actuarially fair contract, he will be indifferent between full coverage and no insurance
 - $EV(\text{fair insurance}) = EV(\text{no insurance})$
 - ➔ $U(\text{fair insurance}) = EU(\text{no insurance})$

- Risk loving and risk neutral agents would not buy full coverage if the premium is actuarially unfair.
 - $EV(\text{unfair insurance}) < EV(\text{no insurance})$
 - ➔ $U(\text{unfair insurance}) < EU(\text{no insurance})$

Risk-averse agents

- If a risk averse agent is offered an actuarially fair contract, he will choose full coverage
 - The actuarially fair contract gives him his expected wealth for sure:
 - $EV(\text{fair insurance}) = EV(\text{no insurance})$
 - ➔ $U(\text{fair insurance}) > EU(\text{no insurance})$
- If a risk averse agent is offered an actuarially unfair contract, the maximum premium he is willing to pay is such that, once paid, he is left with the certainty equivalent of the lottery: $P \leq W - CE$
 - If $P = W - CE$: $EU(\text{no insurance}) = U(\text{unfair insurance})$
- Summing up:
 - An actuarially fair insurance leaves a risk averse agent with the EV for sure
 - A less than actuarially fair insurance must leave the agent with **at least** the CE for sure

Risk-averse agents



Summing up

- With full coverage, and with actuarially fair insurance
 - risk-averse agents would buy the insurance
 - Risk neutral agents would be indifferent to buy the insurance at this price
 - Risk loving agents would not buy it
- With full coverage, and with less than actuarially fair insurance
 - risk-averse agents would buy the insurance paying a **maximum premium** equal to **the value of the object *minus* the certainty equivalent** (in order to obtain at least the same expected utility as without insurance)
 - Risk neutral and risk lovers would not buy the insurance